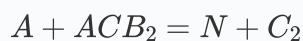
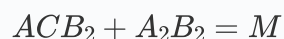
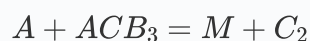


Question

A, **B**, **C** represent three common elements, while **M** and **N** are two compounds that can be considered the same type of substance under the classification of "acid, base, salt, oxide, or elemental substance." They conform to the following chemical reaction processes:



These equations are not balanced. Please provide the chemical formulas of **M** and **N**.

A. M: H_2CO_3 ,
N: CO_2

B. M: Na_2CO_3 ,
N: $NaHCO_3$

C. M: Na_3NO_4 ,
N: Na_3NO_3

D. M: Na_2SO_4 ,

N: Na_2SO_3

E. M: H_2SO_4 ,
N: SO_2

F. M: CO_2 ,
N: CO

G. M: Na_3SO_4 ,
N: Na_3SO_3

H. M: $\text{Na}_2\text{S}_2\text{O}_4$,
N: Na_2SO_3

I. M: $\text{Na}_2\text{N}_2\text{O}_4$,
N: Na_2NO_3

J. M: Na_2CO_3 ,
N: $\text{Na}_2\text{C}_2\text{O}_4$

Answer

Correct Answer: **C**

Detailed Explanation

First, the obtained chemical formulas should be reasonable existing forms. Option G contains Na_3SO_4 , which does not exist, and Option I contains Na_2NO_3 , which does not exist. Therefore, Options G and I are incorrect.

CHECKPOINT

1 PTS

Option G contains Na_3SO_4 , which does not exist, and Option I contains Na_2NO_3 , which does not exist. Therefore, Options G and I are incorrect.

Given that **M** and **N** in the question are two compounds, they can be considered as the same type of substances under the classification of "acid, base, salt, oxide, or simple substance." Thus, Options A and E can be excluded.

CHECKPOINT

1 PTS

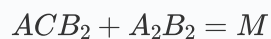
In Options A and E, M and N belong to acid and oxide, respectively, and are not the same type of substances. Therefore, Options A and E are excluded.

According to the addition reaction equation, M must contain all three elements A, B, and C, and the number of atoms of A and B in M must be greater than that of C. Option F contains only two elements and is incorrect. Option H does not conform to the atomic count rule and is also incorrect.

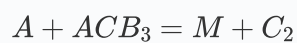
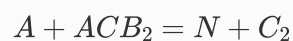
CHECKPOINT

1 PTS

According to the addition reaction equation, M must contain all three elements A, B, and C, and the number of atoms of A and B in M must be greater than that of C. Option F contains only two elements and is incorrect. Option H does not conform to the atomic count rule and is also incorrect.



If the coefficients of the reactants are all 1, M can be derived as A_3CB_4 . Among the options, Option C corresponds to this formula, which, although uncommon, exists. Substituting **A**=Na, **B**=O, and **C**=N into the remaining equations:



From the equations, N can contain at most three elements, so Option B is incorrect.

CHECKPOINT

0.5 PTS

N can contain at most three elements, so Option B is incorrect.

The generated N: Na_3NO_3 is uncommon but exists, and the production of N_2 is reasonable. The other chemical formulas are also reasonable, so Option C is correct.

Specifically: M: Na_3NO_4 , N: Na_3NO_3

CHECKPOINT

1 PTS

M: Na_3NO_4 , N: Na_3NO_3

For Options D and J, A is Na , B is S or C , and C is O . Substituting these into the chemical formulas reveals the existence of unreasonable substances such as NaCO_2 and NaSO_3 . Therefore, Options D and J are incorrect.

CHECKPOINT

1 PTS

When A is Na , B is S or C , and C is O , the equations contain unreasonable substances such as NaCO_2 and NaSO_3 . Therefore, Options D and J are incorrect.